BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 032 · Firewalls and Network Security

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How does a brute-force attack work?

Q2. Which medium resists electromagnetic interference?
A. Copper Ethernet cable
B. Wi-Fi radio
C. Bluetooth radio
D. Fibre optic cable

Q3. How does wired Ethernet differ from Wi-Fi?

Q4. Choose links for a nearby wireless keyboard and a high-capacity link between school buildings. Why does Bluetooth suit the keyboard but usually not the buildings?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Vulnerability — Weakness that could be exploited to compromise a system.

Firewall — Monitors/filters network traffic using security rules.

Anti-malware — Software detecting and removing malicious programs.

Security update — Change fixing bugs or security weaknesses.

User access levels — Permissions controlling which resources a user may access.

Least privilege — Giving users only the access needed for their job.

New learning

Vulnerabilities: A vulnerability is a weakness that an attacker might exploit. Using multiple security measures reduces risk.

Firewalls: A firewall monitors/filters network traffic using rules, blocking or allowing connections according to security policy.

Anti-malware and updates: Anti-malware software detects and may remove malicious software. Software updates can fix known security weaknesses.

User access levels: Access rights restrict what users can view or change, reducing the damage possible if a lower-privilege account is compromised.

Worked example

Explain how a firewall and user access levels help protect a school file server. [4 marks]

1. The file server faces network traffic and has confidential resources.

2. A firewall filters incoming/outgoing network traffic [1], blocking connections that violate security rules [1].

3. User access levels limit permissions [1], preventing ordinary users from editing or viewing protected files [1].

4. Two distinct defences with their actions and security benefits.

We Do · Practise with guidance

Guided 1

A classroom PC may be infected by downloaded malware. Explain how anti-malware software and updates reduce risk. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Worked method: Anti-malware scans for malicious code [1] and can quarantine/remove it [1]. Security updates patch known software weaknesses [1], making exploitation of those vulnerabilities less likely [1].

Guided 2

A school wants to limit access to staff records. Explain how access levels can help. [4 marks]

Worked method: Permissions give access only to authorised accounts [1], restricting who can read records [1]. A normal student account has lower privileges [1], reducing the changes it can make to sensitive information [1].

Guided 3

A school firewall blocks an unfamiliar network connection. Explain what the firewall does and one limitation. [4 marks]

Worked method: Firewall uses rules to check network traffic [1] and blocks disallowed connections [1]. It cannot guarantee stopping every threat [1], especially if a user runs malware already inside the device [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the simulated firewall action.

```
blocked = true
if blocked then
    print("Connection blocked")
endif
```

predict

Question 1

Define a firewall. [2 marks] Source: Adapted from your Paper 1 revision booklet, 1.4.2.

Question 2

A public library allows visitors to use shared PCs while storing private customer booking details. Explain how anti-malware software and user access levels can protect the library's systems. [4 marks] Source: Original OCR J277-style question, not a past-paper original.

Question 3

Explain how firewalls, updates and permissions combine to reduce security risks. [6 marks] Source: Original OCR J277-style question, not a past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

Installing one firewall guarantees that a computer cannot be infected by malware.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It tries many possible passwords or keys until one works.

Q2. D — Fibre optic cable

Q3. Ethernet uses cables. Wi-Fi uses radio, allowing movement but risking interference.

Q4. Bluetooth suits a nearby keyboard's short-range, low-data needs. Links between buildings need more range and capacity, such as fibre.

I Do / We Do pages · Guided practice

Q1. Anti-malware scans for malicious code [1] and can quarantine/remove it [1]. Security updates patch known software weaknesses [1], making exploitation of those vulnerabilities less likely [1].

Q2. Permissions give access only to authorised accounts [1], restricting who can read records [1]. A normal student account has lower privileges [1], reducing the changes it can make to sensitive information [1].

Q3. Firewall uses rules to check network traffic [1] and blocks disallowed connections [1]. It cannot guarantee stopping every threat [1], especially if a user runs malware already inside the device [1].

You Do pages · Code First

```
blocked = true
if blocked then
    print("Connection blocked")
endif
```

Expected result: Connection blocked

You Do pages · Independent questions

Q1. Monitors/filters network traffic [1] according to rules to block unauthorised traffic [1].

Q2. Anti-malware software scans for malicious programs [1] and can quarantine/remove them to reduce harm [1]. User access levels restrict permissions

on public accounts [1], preventing visitors from accessing or changing sensitive customer-booking records [1].

Q3. Firewall filters traffic [1], blocking disallowed connections [1]. Updates repair software weaknesses [1], reducing opportunities for exploitation [1]. Access levels restrict users' permissions [1], limiting exposure or damage [1].

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: